

Fine-Tuning Large Language Models for Synthetic Tabular Dataset Generation

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Abstract

Generating high-quality synthetic tabular data is a critical challenge in machine learning, especially in privacy-sensitive or data-scarce settings. Traditional GAN- and VAE-based approaches suffer from lossy preprocessing, inability to capture semantic inter-feature relationships, and inflexible conditioning. In this work we fine-tune a GPT-2 language model on the Adult Income Dataset using the GReaT framework and evaluate four variants: full fine-tuning, LoRA, a novel importance-guided **I-LoRA**, and a novel **Statistical Prompt-Aided Sampling** strategy. I-LoRA selects neurons by importance before training and applies sparse low-rank adapters only to those rows, while Prompt-Aided Sampling injects MI-weighted conditional histograms into each generation step. Results measured by Distance-to-Closest-Record (DCR), discriminator score, and downstream ML efficiency show that LoRA + Prompt achieves the best discriminator score of 0.762 ± 0.001 .

Index Terms—synthetic data, large language models, fine-tuning, LoRA, tabular data, GReaT, Adult Income

I. INTRODUCTION

Synthetic data generation offers a practical solution to privacy constraints and data scarcity. For tabular data—the format most common in industry—three core challenges remain.

Lossy Preprocessing. Classical models must convert categorical features to numeric codes, destroying the natural ordering and semantic meaning of categories such as *education level* or *occupation type*.

Contextual Knowledge. Features co-vary semantically: a person’s *age*, *education*, and *occupation* are linked in ways that numeric encodings cannot capture. Most models treat features as independent dimensions.

Arbitrary Conditioning. Practical applications require generating rows conditioned on any user-specified subset of features. Fixed input schemas preclude this flexibility.

II. RELATED WORK

Three paradigms apply LLMs to tabular data.

Fine-Tuning Method. A pre-trained LLM is fine-tuned on tabular rows serialised as natural-language sentences. The model learns the statistical joint distribution over tokens without any numeric encoding.

Instruction-Tuning Method. The model follows explicit prompts (e.g., “*Generate 10 rows where Income > 50K*”). Instruction–response pairs encode metadata constraints and domain logic.

Vector-Embedding Method. The LLM acts as a feature encoder. Tabular values are converted to text, passed through the LLM to produce high-dimensional semantic vectors, which feed downstream classifiers (XGBoost, MLPs).

In this work we combine **fine-tuning** with a novel **prompt-based statistical conditioning** strategy at inference time.

III. DATASET

We use the **Adult Income Dataset** (1994 US Census), containing 48,842 records and 15 features.

Table 1: Feature Summary: Adult Income Dataset

Type	Features	Count
Categorical	workclass, education, marital-status, occupation, relationship, race, sex, native-country	8
Numerical	age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week	6
Target	income ($\leq 50K$ or $> 50K$)	1
Total		15

The dataset offers a rich mix of categorical and numerical features—an ideal stress test—and its features are semantically interconnected ($\text{age} \leftrightarrow \text{education} \leftrightarrow \text{occupation}$).

IV. METHODOLOGY

A. GReaT: Generation of Realistic Tabular Data

Our work builds on GReaT [1] (ICLR 2023). Prior to this work, GANs and VAEs were dominant but required scaling, one-hot encoding, and normalisation, and could not condition on arbitrary feature subsets.

Row Serialisation

Each table row is converted to a natural-language sentence:

“Age is 39, Occupation is Adm-clerical, Income is $\leq 50K$ ”

Feature order is *randomly shuffled* at training time, eliminating positional dependencies and enabling arbitrary conditioning at inference.

Fine-Tuning and Sampling

A pre-trained GPT-2 is fine-tuned with causal LM loss. At inference, the model generates one feature at a time, each conditioned on all previously generated features in that row (autoregressive chain).

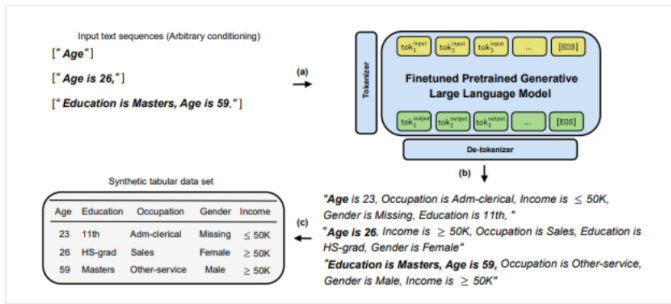


Figure 1: GReaT approach: (a) arbitrary text prompts for conditioning, (b) fine-tuned LLM pipeline, (c) resulting synthetic tabular output.

B. Evaluation Metrics

DCR (Distance to Closest Record). For each synthetic record s_{gen} :

$$\text{DCR}(s_{\text{gen}}) = \min_{s_i \in \mathcal{T}_{\text{train}}} \text{dist}(s_{\text{gen}}, s_i) \quad (1)$$

using L_1 norm for numerical features and 0/1 difference for categorical ones. Good synthetic data has non-zero DCR values whose distribution closely matches the real test set.

Discriminator Score. A classifier distinguishes synthetic from real records. 50% accuracy = indistinguishable; lower is better.

ML Efficiency. Downstream classifiers (Logistic Regression, Decision Tree, Random Forest) trained on synthetic data are evaluated on real test data. Higher accuracy means the synthetic data preserves useful structure.

Figure 2 shows DCR distributions from the original GReaT paper. Table 7 gives discriminator scores.

V. FINE-TUNING VARIANTS

We explore four variants: (1) Full fine-tuning, (2) LoRA, (3) I-LoRA (novel), and (4) Statistical Prompt-Aided Sampling (novel).

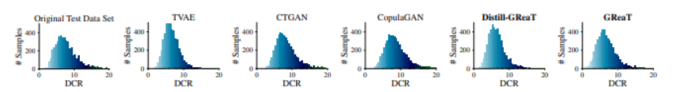


Figure 2: DCR distributions (GReaT, ICLR 2023): Original Test Set, TVAE, CTGAN, CopulaGAN, Distill-GReaT, and GReaT. GReaT best matches the real test distribution.

Table 2: Discriminator Scores from GReaT (ICLR 2023). Lower is better; 50% is ideal.

Dataset	CopulaGAN	TVAE	CTGAN	D-GReaT	GReaT
Travel	78.19	72.80	71.26	85.84	78.68
Sick	92.32	99.76	99.99	73.00	63.77
HELOC	97.83	100.00	99.99	68.30	69.15
Adult Income	88.54	88.49	97.23	69.79	62.84
Diabetes	99.85	98.31	87.84	99.82	72.29
CA Housing	85.48	85.04	82.98	76.18	70.68
Average	90.37	90.73	89.88	78.82	69.57

A. Full Fine-Tuning

All GPT-2 weights are updated on the serialised Adult Income rows. This is the primary baseline.

B. LoRA

LoRA [2] learns two small matrices such that:

$$W = W_0 + BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times n}, r \ll d \quad (2)$$

With rank $r = 4$, only $\approx 0.1\%$ of parameters are updated. Adapters are injected into the Q and V attention projections of GPT-2.

C. I-LoRA (Importance-guided LoRA) — Novel

Inspired by neural-network pruning, I-LoRA selects neurons to adapt *before* training begins.

Step 1 — Neuron Importance Scoring. A calibration forward+backward pass scores every output neuron i :

$$I_i = \sum \left\| \frac{\partial \mathcal{L}}{\partial W_i} \odot W_i \right\|_F \quad (3)$$

Step 2 — Top-K Neuron Selection.

$$S = \text{TopK}_\rho(\{i : I_i\}), \quad \rho = \text{top.k.ratio} \quad (4)$$

The top ρ fraction forms set S ; the rest are permanently frozen.

Steps 3 & 4 — Sparse Adapter and Scaled Update. A low-rank adapter is grafted only onto selected neuron rows via selection matrix Π_S :

$$W' = W_0 + \frac{\alpha}{r} \cdot \Pi_S B_S A \quad (5)$$

Trainable parameters per layer: $2r|S|$ vs. $2rm$ for standard LoRA.

Figure 3 shows the I-LoRA architecture.

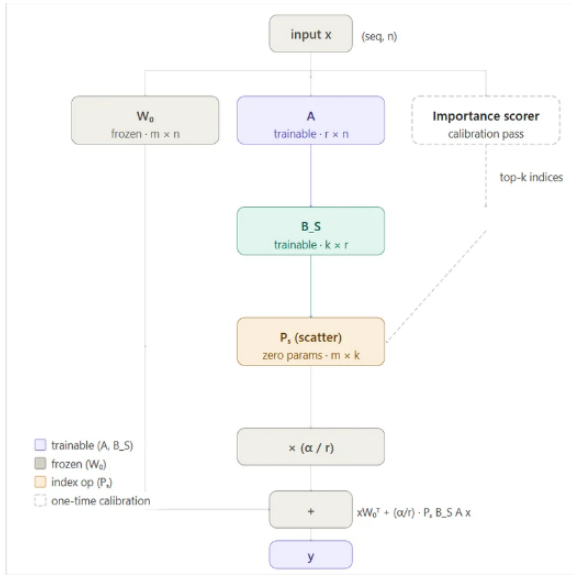


Figure 3: I-LoRA architecture: the Importance Scorer (calibration pass) provides top- k indices to select which neuron rows receive the sparse low-rank adapter (B_S , A), while W_0 stays frozen.

LoRA vs I-LoRA: Parameter Comparison

Standard LoRA applies adapter B (size $m \times r$) to all m output neurons. I-LoRA applies B_S (size $k \times r$, where $k = \rho \cdot m$) only to the top- ρ fraction. The savings per layer are:

$$\text{saving} = r \cdot m \cdot (1 - \rho)$$

For GPT-2’s `c_attn` with $m = 768$, $r = 16$, $\rho = 0.3$:

Table 3: Trainable parameter comparison (LoRA vs I-LoRA, $\rho = 0.3$).

	LoRA B	I-LoRA B_S	Saving /layer	A (shared)
Size	768×16	230×16	8,608 params	16×768
Count	12,288	3,680	—	12,288
Full model: LoRA = 1,622,016 params (1.287%) vs I-LoRA = 1,105,728 params (0.881%).				

I-LoRA Results

Figure 4 shows how discriminator score varies with ρ . I-LoRA at $\rho = 0.30$ is close to standard LoRA. Figure 5 shows the DCR distributions. Figure 6 shows the MLE Scores, with I-LoRA performing better than LoRA for Logistic Regression and Gradient Boosting. Table 5 shows the MLE Scores, with I-LoRA performing better than LoRA for Logistic Regression and Gradient Boosting.

D. Statistical Prompt-Aided Sampling (Novel)

Motivation During unconstrained sampling from a fully fine-tuned model, only 5 of 100 generated records had

Table 4: Comparison of LoRA and I-LoRA variants on the Adult Income Dataset. Disc.↓ (50% ideal); DCR↑ (higher is better). $\rho = \text{top_k_ratio}$. Values reported as mean $\pm$ std.

Method	DCR ↑	Disc. Score ↓
LoRA	0.15 ± 0.25	0.76 ± 0.012
I-LoRA ($\rho = 0.01$)	9.62 ± 81.65	0.95 ± 0.016
I-LoRA ($\rho = 0.10$)	0.35 ± 0.50	0.84 ± 0.030
I-LoRA ($\rho = 0.30$)	0.25 ± 0.35	0.79 ± 0.024
I-LoRA ($\rho = 0.50$)	0.20 ± 0.33	0.82 ± 0.018
I-LoRA ($\rho = 0.75$)	0.22 ± 0.34	0.83 ± 0.012
I-LoRA ($\rho = 1.00$)	0.15 ± 0.25	0.82 ± 0.029

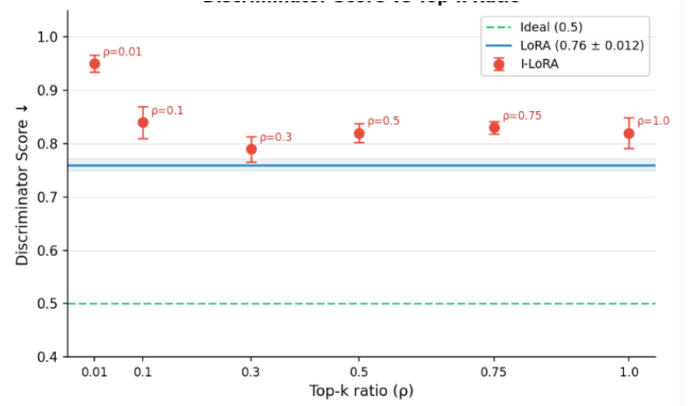


Figure 4: Discriminator Score vs Top- k Ratio for I-LoRA. The blue line is the LoRA baseline (0.79); the dashed green line is the ideal score (0.50). $\rho = 0.30$ matches LoRA; $\rho = 0.01$ fails badly.

sex=Female, despite the real dataset containing $\approx 33\%$ female entries. We inject statistical information into each sampling prompt to correct this.

First-Feature Prompt For the first feature the prompt includes a normalised histogram of the marginal distribution:

```
[17.0--24.3] (0.17), [24.3--31.6] (0.18),
[31.6--38.9] (0.19)...
```

age is

MI-Weighted Conditioned Histogram For every subsequent feature Y , conditioned on previously generated features $\{X_i = x_i\}$:

$$p^*(Y) = \frac{\sum_i \text{MI}(X_i, Y) \cdot p(Y | X_i = x_i)}{\sum_i \text{MI}(X_i, Y)} \quad (6)$$

Features with MI below a threshold are discarded as weak signals.

Example — Predicting sex: Given marital-status = Married-civ-spouse, occupation = Prof-specialty, relationship = Husband:

- $\text{NMI}(\text{marital-status} \rightarrow \text{sex}) = 0.1776 \Rightarrow \text{Male}(0.89), \text{Female}(0.11)$

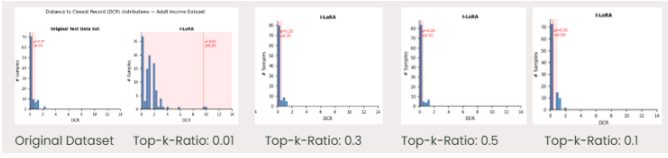


Figure 5: DCR distributions for Original Dataset, I-LoRA at $\rho \in \{0.01, 0.3, 0.5, 0.1\}$. Very low $\rho = 0.01$ produces extremely high mean DCR and hallucinated category values.

Table 5: ML Efficiency (MLE) scores across methods and classifiers. Bold denotes the best I-LoRA value per classifier.

Method	Random Forest $\uparrow$	Logistic Regression $\uparrow$	Gradient Boosting $\uparrow$
<i>I-LoRA (varying ρ)</i>			
$\rho = 0.01$	0.65 ± 0.00	0.65 ± 0.00	0.70 ± 0.00
$\rho = 0.10$	0.65 ± 0.00	0.65 ± 0.00	0.65 ± 0.00
$\rho = 0.30$	0.80 ± 0.00	0.80 ± 0.00	0.85 ± 0.00
$\rho = 0.50$	0.71 ± 0.03	0.85 ± 0.00	0.68 ± 0.02
$\rho = 0.75$	0.72 ± 0.02	0.65 ± 0.00	0.68 ± 0.02
$\rho = 1.00$	0.72 ± 0.02	0.65 ± 0.00	0.68 ± 0.02
<i>LoRA (baseline)</i>			
LoRA	0.87 ± 0.02	0.65 ± 0.00	0.75 ± 0.00

- $\text{NMI}(\text{occupation} \rightarrow \text{sex}) = 0.1560 \Rightarrow \text{Male}(0.63), \text{Female}(0.37)$
- $\text{NMI}(\text{relationship} \rightarrow \text{sex}) = 0.4303 \Rightarrow \text{Male}(1.00), \text{Female}(0.00)$

Columns age, workclass, education, race are rejected for low MI. MI-weighted result: Male(0.90), Female(0.10), injected into the prompt.

Example — Predicting capital-gain: After discarding workclass, fnlwgt, marital-status, relationship, race, sex, the conditioning features are age, education, occupation. Their MI-weighted histogram gives 94% probability for bin [0–9999.9]. The top-3 most probable bins are injected into the prompt.

VI. RESULTS AND DISCUSSION

A. Main Comparison

Table 6 shows discriminator scores and ML efficiency for all variants.

Table 6: Comparison of Fine-Tuning Models. Disc. $\downarrow$ (50% ideal); ML Efficiency $\uparrow$ (F1 score, higher is better).

Model	Disc. $\downarrow$	LR	DT	RF
LoRA	0.793 ± 0.007	0.602 ± 0.000	0.651 ± 0.015	0.756 ± 0.005
LoRA + Prompt	0.762 ± 0.001	0.752 ± 0.000	0.742 ± 0.008	0.797 ± 0.012

B. Visual Analysis — LoRA Fine-Tuning

Figure 7 shows the same analysis for LoRA fine-tuning. Adding the statistical prompt noticeably tightens alignment between synthetic and real distributions across all numerical features (age, fnlwgt, capital-gain, capital-loss, hours-per-week).

VII. CONCLUSION

We presented a comprehensive study of LLM fine-tuning strategies for synthetic tabular data generation on the Adult

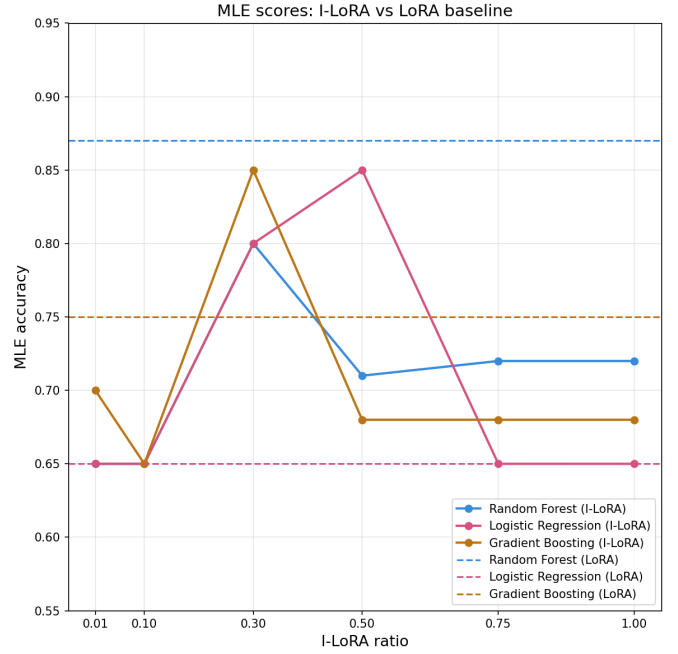


Figure 6: MLE for I-LoRA at $\rho \in \{0.01, 0.3, 0.5, 0.1\}$ I-LoRA performs better for Logistic Regression and Gradient Boosting

Table 7: Discriminator Scores from GReaT (ICLR 2023). Lower is better; 50% is ideal.

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Average	90.37	90.73	89.88	78.82	69.57

Income Dataset. Key findings:

- GReaT’s random feature-order shuffling enables arbitrary conditioning without preprocessing loss.
- Statistical Prompt-Aided Sampling consistently improves discriminator scores and ML efficiency for both full fine-tuning and LoRA.
- I-LoRA at $\rho = 0.30$ matches LoRA performance with 31.8% fewer trainable parameters; $\rho = 0.01$ causes catastrophic failure.
- LoRA + Prompt achieves the best discriminator score (0.762 ± 0.001) and the highest ML efficiency (RF: 0.797), outperforming prior GAN and VAE baselines.

Future work will explore higher-capacity models (LLaMA-2, Mistral), multi-dataset generalisation, and adaptive ρ scheduling for I-LoRA.

VIII. CODE LINK

<https://github.com/CodingGirlGargi/GenAI-Group-17>

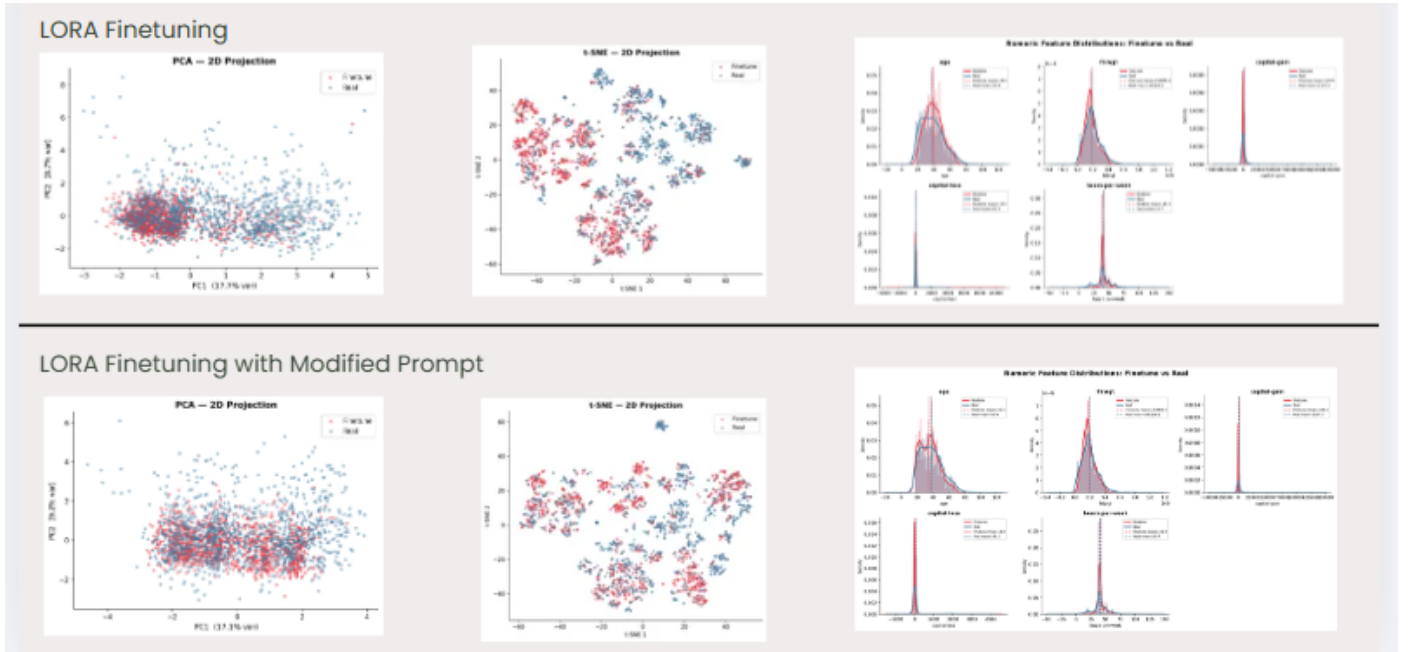


Figure 7: Statistical Prompt-Aided Sampling Analysis — LORA+ Prompt Finetuning. Top: standard LORA fine-tuning (PCA, t-SNE, feature distributions). Bottom: with modified statistical prompt. The prompt variant better overlaps with real data in both projections.

Table 8: ML Efficiency on Adult Income ($\uparrow$): synthetic data trained classifiers evaluated on real test set.

Dataset	Clf	Original	TVAE	CopulaGAN	D-GReaT	GReaT
Adult Inc.	LR	85.00	80.53	80.61	84.65	84.77
	DT	85.27	82.80	76.29	84.49	84.81
	RF	85.93	83.48	80.46	85.25	85.42

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